frontier is also not bullet shaped. Constraints inhibit what an investor can do and an investor can only be made (weakly) worse off. If an investor is lucky, the best risk–return trade-off is unaffected by adding constraints. We see this in Figure 3.7 in the region where the constrained and the unconstrained mean-variance frontiers lie on top of each other. But generally constraints cause an investor to achieve a worse risk–return trade-off. Nevertheless, even with constraints, the concept of diversification holds: the investor can reduce risk by holding a portfolio of assets rather than a single asset.

2.5. THE RISKS OF NOT DIVERSIFYING

Many people hold a lot of stock in their employer. Poterba (2003) reports that for large defined-contribution pension plans, the share of assets in own-company stock is around 40%. This comes usually, but not always, from discounted purchases and is designed by companies to encourage employee loyalty. For individuals, such concentrated portfolios can be disastrous—as employees of Enron (bankrupt in 2001), Lucent (which spiraled downward after it was spun off from AT&T in 1996 and was bought for a pittance by Alcatel in 2006), and Lehman Brothers (bankrupt in 2008) found out. Enron employees had over 60% of their retirement assets in company stock when Enron failed.[8] According to the Employee Benefits Research Institute, among 401(k) plan participants who had equity exposure in 2009, 12% had company stock as their only equity investment. For equity market participants in their sixties, 17% had equity exposure to equity markets only through their employers' stock.[9]

While the cost of not diversifying becomes painfully clear when your company goes bankrupt, mean-variance investing reveals that there is a loss even when your employer remains solvent. Individuals can generate a higher risk–return trade-off by moving to a diversified portfolio. Poterba (2003) computes the cost of not diversifying a retirement account relative to simply investing in the diversified S&P 500 portfolio. Assuming that half of an individual's assets are invested in company stock, the certainty equivalent cost (see chapter 3) of this concentrated position is about 80% of the value of investing in the diversified S&P 500 portfolio. This is a substantial reduction in utility for investors. An individual should regularly cash out company stock, especially if the stock is rising faster than other assets in his portfolio. As own-company stock rises relative to other assets, it represents even greater concentrated risk for that investor.[10] (Furthermore, your human capital itself is concentrated with that employer, see chapter 5.)

[8] See Barber and Odean (2011).

[9] See Van Derhei, J., S. Holden, and L. Alonso, 2010, 401(k) Plan Asset Allocation, Account Balances and Loan Activity in 2009, Employee Benefit Research Institute Issue Brief No. 350.

[10] Individual investors, unfortunately, tend to do exactly the opposite—individuals hold larger amounts in employer stocks, which have had the strongest return performance over the last ten years, as Benartzi (2001) shows.